

PROJECT Proposal

Title of the Project: Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains

Problem Statement (SIH26028):

Indian Railways operates thousands of coaching trains daily across an interconnected network. Existing systems rely on static timetables and simple station-by-station estimates that fail to accurately predict downstream arrivals when delays occur. This project aims to build an intelligent, data-driven system for the dynamic forecasting of Expected Time of Arrival (ETA) for coaching trains, leveraging real-time data inputs such as train movement, section congestion, weather conditions, and train priority schedules to dynamically adjust and recalculate ETAs in real-time.

Technical details of the Projects:

- **Motivation of the Project:** Current arrival forecasting mechanisms fail to account for non-linear cascading delays, track bottlenecks, weather visibility issues, and train signal priority conflicts. By shifting from static schedule extrapolation to a dynamic machine learning pipeline, the platform will deliver highly accurate downstream arrival predictions, optimizing platform allocation, terminal management, and passenger communication.

This project is proposed to address these fundamental challenges by building a centralized, intelligent prediction platform. By combining live GPS telemetry streams with spatial-temporal graph neural models and sequential time-series algorithms, the system dynamically accounts for network topology, track capacity utilization, weather visibility, and priority rules.

Work Plan:

- **Methodology:** The project will follow a modular application-development approach consisting of three major components: a real-time data ingestion and processing pipeline, a hybrid machine learning prediction engine, and an administrative web dashboard for operational analytics.

The system will be developed using React with Tailwind CSS for the administrative and operations dashboard, Python (FastAPI/Node.js) for the backend services, and PyTorch / XGBoost for the machine learning forecasting models. The backend will manage live data ingestion, spatial graph construction, historical log processing, contextual feature engineering, dynamic model inference, WebSocket distribution, and alert notifications.

The development process will be divided into the following stages:

1. Requirement analysis and system planning.
2. Database and backend architecture design.
3. Data ingestion and real-time tracking stream parser module.

4. Spatial-temporal railway network graph construction.
5. Feature engineering layer (train priority, section capacity, and weather metrics integration).
6. Machine learning ETA forecasting model architecture and baseline training.
7. Dynamic inference pipeline and continuous retraining workflow.
8. Real-time WebSocket event streaming and API integration.
9. Administrative operations dashboard and live map visualization.
10. Delay heatmap generation and station platform schedule prediction.
11. Anomaly detection and automated alert notification workflow.
12. System monitoring, model evaluation (MAE/RMSE validation), and audit logging.
13. Integration testing and final system performance validation.

Organization of work element:

- **Data Engineering & Ingestion Pipeline:** Collection, cleaning, and spatial mapping of live coordinates and station network graphs.
- **AI/ML Delay Prediction Core:** Model architecture, feature engineering, and real-time inference logic.
- **Backend API & Web Services:** Server implementation, WebSocket management, and database synchronization.
- **Frontend Visualization Client:** Interactive map UI, train search, and platform allocation analytics dashboard.

Time Schedule:

- Start date : 12 September 2026
- End date : Ongoing

Proposed outcome/ findings :

The proposed outcome of the project is a functional digital platform that provides an organized workflow for dynamic railway arrival forecasting and operational management.

The completed system is expected to provide:

1. Real-time ingestion and cleaning of continuous train telemetry streams.
2. Spatial-temporal graph representation of railway tracks, junction nodes, and station hubs.
3. Dynamic feature integration including train priority class, weather visibility, and section utilization.
4. AI-driven machine learning core for continuous prediction of downstream station arrival times.
5. Automated calculation and minimization of prediction error (Mean Absolute Error).
6. Real-time operations dashboard with interactive track visualization and map rendering.
7. Live section delay heatmaps and congestion tracking across rail corridors.
8. Platform arrival and track allocation forecasting for station masters.
9. Automated alert distribution for severe cascading delay forecasts.
10. Historical audit trails and analytics logs for train performance evaluation.
11. RESTful and WebSocket API endpoints for seamless integration with existing rail passenger display systems.

The project will demonstrate how a centralized software platform can organize the complete lifecycle of dynamic transportation forecasting, from telemetry ingestion and AI inference to operational analytics and passenger communication.

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